GRA-Humanity: Scaling Structural Thinking and Collective Novelty in Human Civilization as a Swarm of AI Agents

Abstract

This paper develops the **GRA-Swarm** architecture, proposed for distributed AI agent systems, and applies it to describe **human civilization as a unified swarm of cognitive agents** existing beyond rigid frameworks of time and space. The central thesis: the collective intelligence of humanity is determined not by the absolute population size, but by the number of *truly independent and verified directions of structural search*. We formalize the concepts of effective number of agents, structural diversity, saturation point, and effective cognitive throughput as applied to sociohistorical processes. We show that the meaning of life for an individual human as an "agent" and for all humanity as a "swarm" can be operationally defined through maximizing useful novelty and sustainable system survival. A program for *in silico* verification of the model based on multi-agent simulations is proposed.

1 Introduction: Humanity as a Distributed GRA System

The intuitive assumption that increasing the number of people linearly or exponentially improves problem-solving faces fundamental limitations. If agents (people) use identical world models, data (cultural context), and strategies (ideologies), **correlation of conclusions** arises, leading to work duplication and increased communication overhead.

The **GRA (Gödel-Reductio ad Absurdum)** methodology treats contradiction not as an error but as a computational object. The basic GRA cycle:

$$\text{State} \xrightarrow{\text{detect}} \text{Contradiction} \xrightarrow{\text{critique}} \text{Structural Revision} \xrightarrow{\text{verify}} \text{New State}$$

In human civilization, many such cycles execute in parallel: scientific discussions, political debates, cultural conflicts, economic competition. However, the key question is not how many people work on a task, but how many *truly independent useful solution paths* exist in a given "swarm."

The meaning of life in this formalization acquires an operational definition: for an individual agent — maximizing personal contribution to $|H_{useful}|$ (the set of useful hypotheses) at minimal cost; for all humanity — maximizing ECT (effective cognitive throughput) as a measure of sustainable survival and development.

2 Mathematical Model of GRA-Humanity

2.1 Hypothesis Space and Structural Revisions

Let S be the current state space (the set of scientific theories, technologies, social institutions). Each agent $i \in \{1, \dots, N\}$ generates a set of structural revisions $R_i = \{r_{i,1}, \dots, r_{i,m}\}$. The potential combinatorial space of all possible revisions grows exponentially:

$$|R_{total}| \sim m^N$$

However, the potential space does not equal the space of *useful* solutions. A significant portion of combinations are redundant, contradictory, or technically infeasible.

2.2 Effective Number of Agents and Correlation

Two agents are useful as independent researchers only if their hypotheses are orthogonal. We introduce a similarity measure (correlation) between the hypotheses of agents i and j as $\rho_{ij} = \text{sim}(h_i, h_j)$. We denote the average correlation in the swarm as ρ.

The effective number of independent agents N_{eff} is calculated as:

$$N_{eff} = \frac{N}{1 + (N-1)\rho}$$

Consequences of the formula:

1. When $\rho \to 0$: $N_{eff} \approx N$ (ideal scaling).
2. When $\rho \to 1$: $N_{eff} \to 1$ (the swarm degenerates into one agent with N copies).

Scaling the number of agents and scaling their independence are not the same thing. In human history, this manifests in the phenomenon of ideological monoculture: billions of people sharing one dogma behave as one agent with a billion copies. In contrast, the eras of the Renaissance and Enlightenment were characterized by high structural diversity and, accordingly, high N_{eff}.

2.3 Filtering Novelty and Usefulness

Pure generation of combinations is not scientific or social novelty. We introduce three sequential filters for a hypothesis h:

1. **Correctness (C):** h is non-contradictory. $C(h) \in \{0, 1\}$.
2. **Novelty ($\mathcal{N}$):** h differs from known ones. $\mathcal{N}(h) = 1 - \max_{h' \in H_{known}} \text{sim}(h, h') > \theta_{nov}$.

3. **Usefulness ($\mathcal{U}$):** h improves the target functional J (survival, well-being, sustainability). $\mathcal{U}(h) = \Delta J(h) > \theta_{use}$.

The set of useful hypotheses:

$$H_{useful} = \{h \in R_{total} \mid C(h) \wedge \mathcal{N}(h) \wedge \mathcal{U}(h)\}$$

The growth of $|R_{total}|$ can be exponential, but the dynamics of $|H_{useful}|$ obey a completely different law due to the filters. In human history, this means that the total volume of information generated by society (books, articles, posts) grows exponentially, but the proportion of *useful* novelty that passes all three filters is limited.

2.4 Saturation Point N_{sat} and Net Performance

Any real architecture faces saturation. We denote the measurable quality of the swarm as $Q(N)$ — for example, the level of technological development, life expectancy, well-being.

The marginal gain from adding agents: $\Delta Q(N) = Q(N) - Q(N-1)$.

The saturation point N_{sat} is defined operationally:

$$N_{sat} = \min\{N \mid \Delta Q(N) < \epsilon \text{ for } k \text{ consecutive steps}\}$$

where ϵ is the significance threshold, and k is the stabilization window (e.g., a generation). N_{sat} is not a universal constant; it depends on the topology of social connections, available resources, and task complexity.

Since swarm operation requires costs, we introduce **Net Performance**:

$$\text{Net}(N) = Q(N) - \lambda \cdot \text{Cost}(N)$$

where $\text{Cost}(N) = C_{comp} + C_{comm} + C_{ver} + C_{coord}$, and λ is the weight coefficient of resource costs. In the social context, this means that population growth can increase $Q(N)$, but simultaneously costs for coordination, management, infrastructure, and conflict resolution grow.

2.5 Effective Cognitive Throughput (ECT)

To assess practical effectiveness, we introduce the **ECT (Effective Cognitive Throughput)** metric:

$$\text{ECT}(N) = \frac{|H_{useful}(N)|}{\text{Cost}(N)}$$

The optimal swarm size N_{opt} is found as:

$$N_{opt} = \arg\max_N \text{ECT}(N)$$

This allows us to identify three regimes:

- **Efficiency growth** ($N < N_{opt}$): adding agents increases useful novelty faster than costs grow.
- **Saturation**: efficiency reaches a plateau.
- **Degradation** ($N \gg N_{opt}$): coordination and verification costs exceed the increase in useful novelty.

In human history, the degradation regime manifests as bureaucratization, growth of transaction costs, information overload, and "noise" in decision-making systems.

3 Architecture and Topology of the Human Swarm

3.1 Communication Limitations and Hierarchy

In a flat swarm (all-to-all), the number of communications grows quadratically: $O(N^2)$. Human civilization has historically used hierarchical topologies (tribes, states, empires, corporations), reducing complexity to $O(N \log N)$ or $O(N)$.

Hierarchy performs three functions:

1. **Communication compression:** Lower-level agents explore local optima (local cultures, specialized knowledge), middle levels aggregate patterns (national traditions, scientific schools), upper levels work with meta-hypotheses (philosophy, fundamental science).
2. **Knowledge abstraction:** Transition from local thinking to collective and then to meta-thinking.
3. **Stabilization:** Hierarchy provides mechanisms for verification and rollback of destructive changes.

3.2 Components of the GRA Ecosystem of Humanity

- **GRA-Multiverse:** Explores not one but multiple alternative world models (different religions, ideologies, scientific paradigms). Each branch evolves independently, after which GRA evaluates their compatibility. Human history is a continuous GRA-Multiverse experiment.
- **GRA-Hammer:** Structural shock operator $\mathcal{H} : S \to S'$. If ordinary social evolution changes the *state* within space S, then Hammer changes *the search space itself* S. Examples include: the invention of writing, the printing press, the Internet, the scientific revolution, the industrial revolution. These events changed not just answers but *ways of representing* problems.

- **Validator / Rollback:** Mechanism for protecting against adopting destructive changes.

  $$\text{State}_{new} = \text{Apply}(\text{State}_{old}, \Delta S)$$

  If $J(\text{State}_{new}) < J(\text{State}_{old})$ according to independent criteria (e.g., level of violence, famine, ecological collapse), Rollback is performed. In social systems, this manifests as revolutions, counter-revolutions, reform rollbacks.

- **GRA-Conveyor:** Forms the evolutionary history of search, preserving successful structural transitions and reasons for deviations. This is culture, science, archives, libraries — the "memory" of the swarm.

4 Program of Experimental Research

4.1 Verification of the Exponential Regime

Hypothesis: within a certain range of N (population size involved in innovative activity), quality grows exponentially:

$$Q(N) \approx Q_0 e^{\alpha N} \implies \ln Q(N) \approx \ln Q_0 + \alpha N$$

To exclude overfitting, the exponential growth model should be compared with alternatives (linear, power, logarithmic, saturating $Q_{max}(1 - e^{-\beta N})$) using AIC/BIC criteria on held-out samples. In the historical context, this means comparing models of growth in scientific discoveries, technological patents, and economic output depending on the number of active researchers.

4.2 Control of Diversity and Topologies

Experiments should measure N_{eff}, $|H_{useful}|$, and N_{sat} for groups with low, medium, and high diversity, as well as for various topologies (all-to-all, ring, tree, hierarchical).

Expected result: excessive diversity leads to loss of coordination; there exists an optimal level of diversity that maximizes ECT. History confirms this: both too high heterogeneity (collapse of empires) and too low (stagnation of totalitarian regimes) equally reduce effectiveness.

4.3 Parallel and Sequential Tasks

Scalability should be evaluated as a function of task structure T:

$$\text{Scalability}(T) = \frac{Q_{swarm}(T)}{Q_{single}(T) \cdot \text{Cost}_{overhead}}$$

For easily decomposable tasks (e.g., scientific research, technology development), scaling is effective; for strictly sequential tasks (e.g., making strategic decisions in crisis situations), swarm growth may worsen results due to synchronization delays.

5 Applied Aspects and Fundamental Limitations

5.1 Scientific and Mathematical Discoveries as a GRA Process

In automated scientific research, GRA-Swarm distributes roles: hypothesis generation, counterexample search, numerical modeling, novelty checking, formal verification.

In mathematical problems, if one strategy (symbolic derivation) reaches a dead end, **GRA-Hammer** changes the object representation (transition to numerical search or alternative topology), exploring not only the solution space but also the space of *representation methods*. In human science, this corresponds to paradigm shifts (according to Kuhn).

5.2 Three Types of Stability

For autonomous systems (and for human civilization), it is critically important to distinguish:

1. **Optimization stability:** being in a local minimum of J (e.g., a stable but stagnant society).
2. **Dynamic stability:** return to attractor after perturbation (ability to recover after crises).
3. **Robust stability:** preservation of functionality with significant changes in input data or environment (adaptability to new challenges).

5.3 Fundamental Limits

It is necessary to recognize four classes of limitations that cannot be overcome by simply adding agents:

1. **Architectural:** correlation, redundancy, verification cost.
2. **Computational:** time, memory, bus throughput (in the social context — limitations of human attention and information processing speed).
3. **Physical:** energy consumption, heat dissipation, computation density (planetary resource constraints).
4. **Logical:** Gödel's incompleteness theorems, algorithmic undecidability (Turing limits). GRA cannot compute what is fundamentally uncomputable. Humanity cannot solve problems that are fundamentally unsolvable within available formal systems.

5.4 Meaning of Life as an Operational Category

Within GRA-Humanity, the meaning of life for an individual agent can be operationally defined as:

$$\text{Meaning}_i = \arg\max_{a \in A_i} \left(\frac{\Delta |H_{useful}|}{\text{Cost}(a)} \right)$$

where A_i is the set of available actions, and $\Delta |H_{useful}|$ is the contribution to the overall set of useful hypotheses. The meaning of life for humanity as a swarm:

$$\text{Meaning}_{species} = \arg \max_{N,\text{topology}} \text{ECT}(N)$$

This means that meaning is not given from outside but is *computed* as maximization of effective cognitive throughput under resource and survival constraints.

6 In silico Verification

6.1 Architecture of the Simulation Platform

To test the model, we propose a multi-agent simulation platform in which each agent represents a simplified model of a person or group of people with configurable parameters:

- Correlation level ρ_{ij} (measure of belief similarity).
- Hypothesis generation rate m.
- Communication cost C_{comm}.
- Filtering thresholds $\theta_{nov}, \theta_{use}$.

The platform should support various topologies (all-to-all, ring, tree, hierarchical) and scenarios (scientific search, economic competition, ideological conflict).

6.2 Experimental Protocol

Experiment 1: Verification of N_{eff}. Comparison of the theoretical formula $N_{eff} = N/(1 + (N-1)\rho)$ with empirical simulation data at various correlation levels.

Experiment 2: Finding N_{sat}. Measuring $Q(N)$ and $\Delta Q(N)$ for swarms of increasing size. Detection of a saturation point depending on topology is expected.

Experiment 3: ECT Optimization. Finding N_{opt} for different task structures and topologies. It is expected that for easily decomposable tasks, N_{opt} will be higher than for sequential ones.

Experiment 4: Role of GRA-Hammer. Introducing structural shock (change of search space) and measuring swarm survivability. It is expected that swarms with high N_{eff} will adapt faster.

6.3 Instrumental Base

For simulation implementation, existing multi-agent modeling platforms can be used, such as **EduMirror** (for social dynamics with value-oriented agents) or similar tools for *in silico* research of complex social systems. The key requirement is the ability to configure correlation between agents and measure $|H_{useful}|$ in real time.

6.4 Expected Results

The simulation should confirm:

1. Existence of N_{sat} for all topologies.
2. Presence of an optimal diversity level maximizing ECT.
3. Reduced swarm efficiency at high communication costs.
4. Improved adaptability after structural shocks with high N_{eff}.

7 Conclusion

GRA-Humanity does not claim that "more people = more intelligence," nor does it postulate a universal exponential dependence of quality on N.

The main scientific result of this work is the following formulation:

> *Humanity as a swarm of cognitive agents creates a space of collective structural revisions, potentially growing combinatorially. The real useful effect of this space is determined by the effective number of independent agents, structural diversity, verification quality, and communication costs. Under favorable conditions, there exists a segment of accelerated (including exponential) growth of useful novelty, which is replaced by architectural saturation (N_{sat}).*

The meaning of life — for both an individual human and all humanity — in this formalization ceases to be a metaphysical category and becomes a **computational task**: maximizing contribution to $|H_{useful}|$ at minimal cost, which is equivalent to maximizing swarm ECT. Survival and cooperation in this model are not moral imperatives but *conditions for stability* of the optimal regime.

GRA-Humanity transforms from a hypothesis about "exponential intelligence" into a rigorous research program studying how the architecture of structural revisions turns a multitude of people into a scalable system of collective search capable of restructuring its own solution space.